142544 – Nyabuto Lorna Njeri 4C
IoT Exe 2
PART A

CIRCUIT 1

1. TR = 20 + 30 + 50 = 100ohms

2. TC = I = V/R = 125/100
   = 1.25A

3. Current through each resistor
   $I_{20} = I_{30} = I_{50} = 1.25$ A

4. Voltage drop across each resistor
   V=IR
   20ohm = 1.25 x 20 = 25V

   30ohm = 1.25 x 30 = 37.5V

   50ohm = 1.25 x 30 = 62.5V

5. Power dissipated
   P = VI
   20ohm resistor = 25 x 1.25 = 31.25W
   30ohm resistor = 37.5 x 1.25 = 46.875W
   50ohm resistor = 62.5 x 1.25 = 78.125W

CIRCUIT 2

1. TR = (1/100) + (1/50) = (1/100) + (2/100) = 3/100 =

   33.33ohms

   R_total = 20 + 33.33 = 53.33 ohms

2. TC = 125/53.33
   = 2.344A

3. Current through each resistor
   Voltage = 125 – (2.344 x 20) = 125 – 46.875
   =78.125V

I_{100} = 78.125/100
=0.781A
I_{50} = 78.125/50 = 1.563A

4. Voltage drop on each resistor
   20ohm = 2.344 x 20 = 46.875V

   100ohm = 78.125V

   50ohm = 78.125V

5. Power dissipated
   20 ohm = 46.875 x 2.344
   = 109.86W
   100 ohm = 78.125 x 0.781
   = 61.03W
   50 ohm = 78.125 x 1.563
   = 122.07W